

Suppose A satisfies the following:

(7P) For all $k \geq 0$ there exists strings x, y, z such that
 $xyz \in A$, $|y| \geq k$ and for all u, v, w
with $y = uvw$ and $v \neq \epsilon$, there exists
 $i \geq 0$ s.t. $xu v^i w z \notin A$.

Then A is not regular.

$$A = \{ww \mid w \in \{a, b\}^*\}.$$

Claim. A is not regular.

Proof. Consider any $k \geq 0$ and let

$$x = \epsilon, y = a^k, z = ba^kb. \text{ Then } xyz \in A.$$

Consider any split of $y = uvw$ s.t. $v \neq \epsilon$.

Say $y = a^j a^m a^n$ where $k = j + m + n$; $m > 0$.

$$\begin{aligned} \text{Let } i = 2. \text{ Then } xu v^2 w z &= a^j a^m a^m a^n b a^k b \\ &= a^{k+m} b a^k b \end{aligned}$$

Since $m > 0$, $a^{k+m} b a^k b \notin A$.

Thus A is not regular.

Use of closure properties.

$$A = \{ x \in \{a,b\}^* \mid \#a(x) = \#b(x) \}.$$

$\#a(x)$: number of a 's in the string x .

Claim: A is not regular.

Proof. Suppose A is regular.

$$A' = A \cap L(a^*b^*).$$

1) $L(a^*b^*)$ is regular.

2) if A is regular and $L(a^*b^*)$ is regular
then A' is regular. Since regular sets are
closed under intersection.

$$A' = A \cap L(a^*b^*) = \quad ?$$

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closed under intersection.

$$A' = A \cap L(a^*b^*) = \underbrace{\{a^n b^n \mid n \geq 0\}}_{\text{is not regular}}.$$

This results in a contradiction

Thus A is not regular.

For a string $x = a_1 a_2 a_3 \dots a_n$, let

$$\text{rev}(x) = a_n a_{n-1} \dots a_2 a_1.$$

$$A \subseteq \Sigma^*, \text{ rev}(A) = \{\text{rev}(x) \mid x \in A\}.$$

Theorem. If A is regular then $\text{rev}(A)$ is also regular.

$$A = \{a^n b^m \mid n \geq m\}$$

Claim. A is not regular.

Proof.

Suppose A is regular. Then $\text{rev}(A)$ is regular.

$$\text{rev}(A) = \{b^m a^n \mid n \geq m\}.$$

Consider the homomorphism h where $h(a) = b$ & $h(b) = a$.

Let $A' = h(\text{rev}(A))$. Since $\text{rev}(A)$ is regular,
 $h(\text{rev}(A))$ is also regular.

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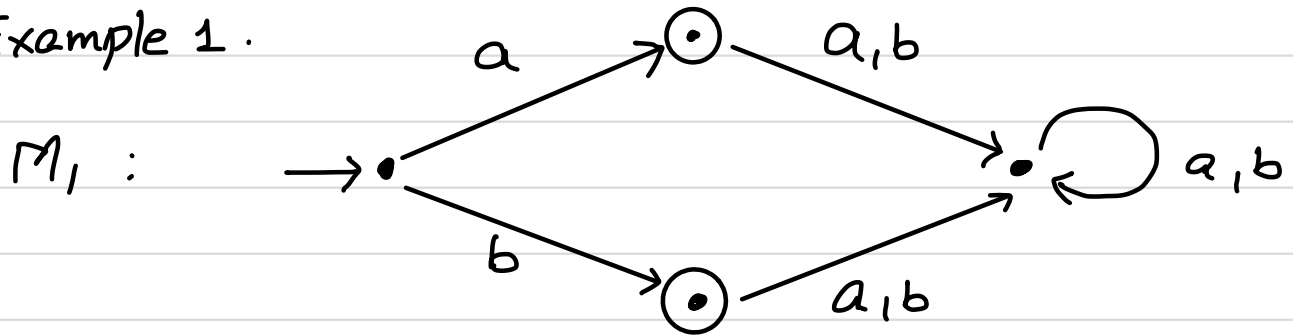
$$A \cap A' = \{ \underbrace{a^n b^n} \mid n \geq 0 \}.$$

is not regular - Contradiction

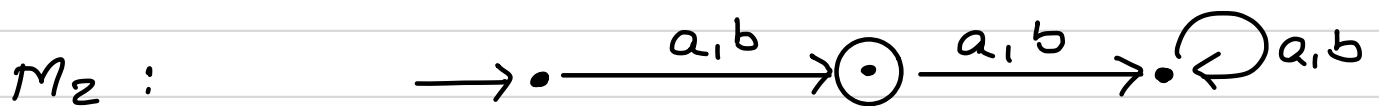
Thus A is not regular.

State Minimization - DFA.

Example 1.



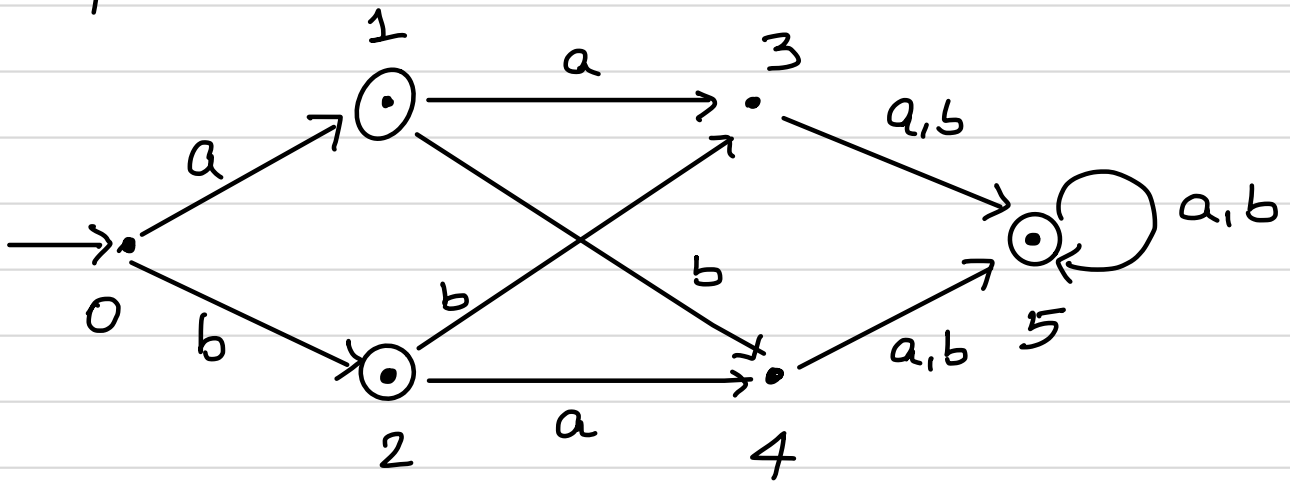
$$L(M_1) = \{a, b\}$$



$$L(M_1) = L(M_2)$$

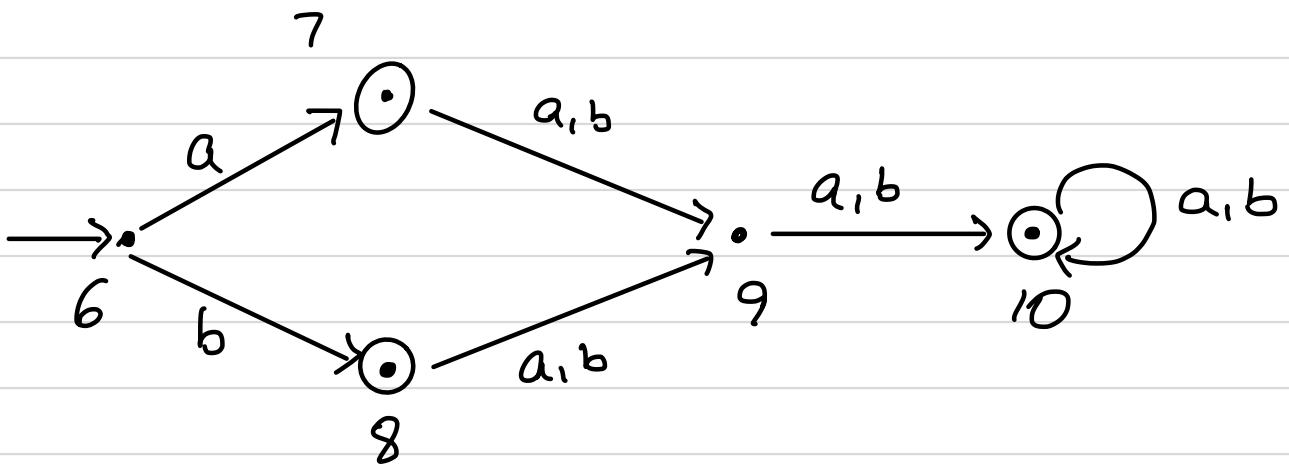
Example 2.

M_1

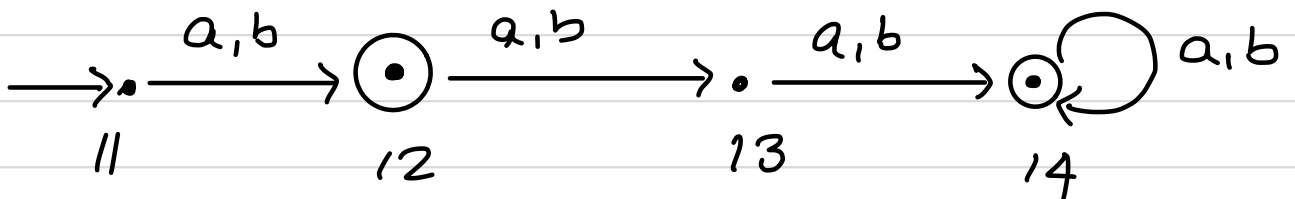


$$L(M_1) = \{a,b\} \cup \{\text{strings of length at least 3}\}$$

M_2

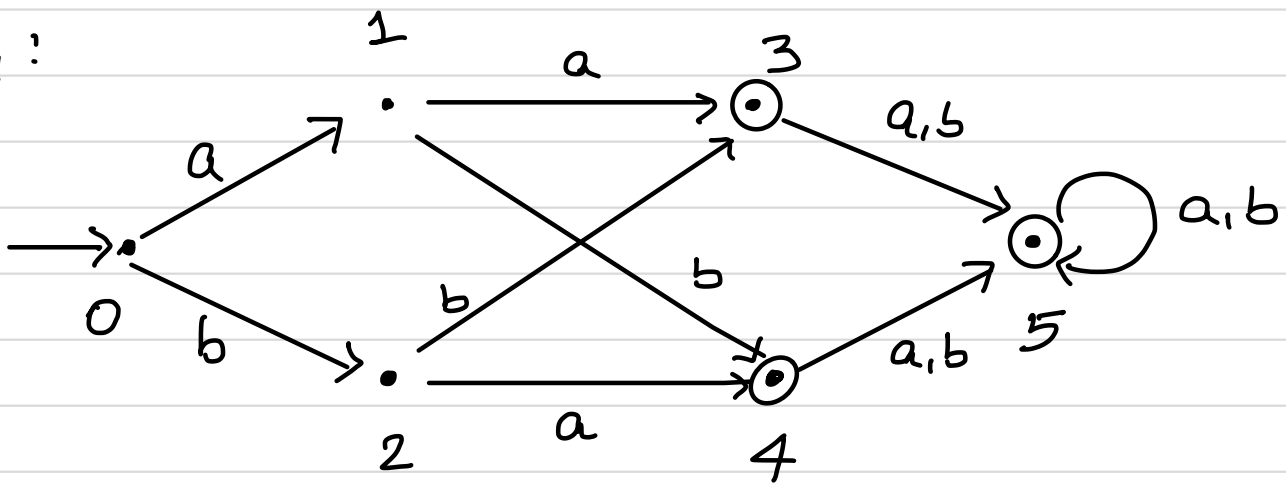


M_3 :



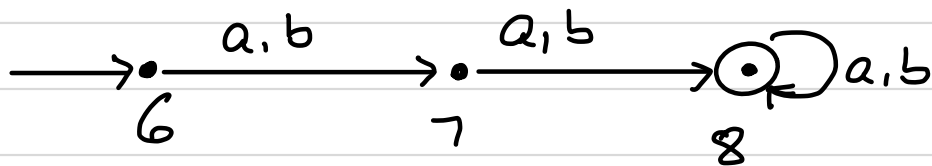
Example 3.

M_1 :



$$L(M_1) = \{ \text{strings of length at least 2} \}$$

M_2



For a DFA $M = (Q, \Sigma, S, \delta, F)$ the state minimization process consists of:

1. Removing inaccessible states.

2. Collapsing "equivalent" states.

- We cannot collapse a final state p and a non-final state q .

if $p = \hat{\delta}(s, x)$ and $q = \hat{\delta}(s, y)$ then
 $x \in L(M)$ and $y \notin L(M)$.

- if we collapse states p & q then we

should also collapse $\delta(p, a)$ and $\delta(q, a)$

Otherwise the resulting automata will not be deterministic.

Observation. Cannot collapse p and q if $\hat{\delta}(p, x) \in F$ and $\hat{\delta}(q, x) \notin F$ for some $x \in \Sigma^*$

Quotient Automata.

$$\text{DFA } M = (Q, \Sigma, \delta, s, F) \quad M/\sim = (Q', \Sigma, \delta', s', F')$$

$Q' = \{[p] \mid p \in Q\}$ - states of $M/\sim$ are the $\sim$ -equivalence classes.

$$\delta'([p], a) = [\delta(p, a)] \quad \delta': Q' \times \Sigma \rightarrow Q'$$

$$s' = [s], \quad F' = \{[p] \mid p \in F\}$$

To show. δ' is well defined.

Quotient Automata.

$$\text{DFA } M = (Q, \Sigma, \delta, s, F) \quad M/\approx = (Q', \Sigma, \delta', s', F')$$

$Q' = \{[p] \mid p \in Q\}$ - states of $M/\approx$ are the $\approx$ -equivalence classes.

$$\delta'([p], a) = [\delta(p, a)]$$

$$s' = [s], \quad F' = \{[p] \mid p \in F\}$$

To show. δ' is well defined.

Lemma 1. if $p \approx q$ then $\delta(p, a) \approx \delta(q, a)$

That is, if $[p] = [q]$ then $[\delta(p, a)] = [\delta(q, a)]$

Proof. Suppose $p \approx q$. For $a \in \Sigma$ and $y \in \Sigma^*$

$$\hat{\delta}(\delta(p, a), y) \in F \text{ iff } \hat{\delta}(p, ay) \in F$$

$$\text{iff } \hat{\delta}(q, ay) \in F \quad [\text{since } p \approx q]$$

$$\text{iff } \hat{\delta}(\delta(q, a), y) \in F.$$

Since the above holds for all $y \in \Sigma^*$, $\delta(p, a) \approx \delta(q, a)$
[By defn of $\approx$].

DFA $M = (Q, \Sigma, \delta, s, F)$ $M/\approx = (Q', \Sigma, \delta', s', F')$

Lemma 2. $p \in F$ iff $[p] \in F'$.

Proof. $\Rightarrow$ Follows from the definition of F'

$\Leftarrow$ if $p \approx q$ and $p \in F$ then $q \in F$. That is,

every $\approx$ -equivalence class is either a subset of F
or disjoint from F .

Follows by taking $x = \epsilon$ in the definition of $p \approx q$.

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Lemma 3. For all $x \in \Sigma^*$, $\hat{s}'([p], x) = [\hat{s}(p, x)]$

Proof. By induction on $|x|$.

Base case: $x = \epsilon$. $\hat{s}'([p], \epsilon) = [p] = [\hat{s}(p, \epsilon)]$

Induction Step.

$$\begin{aligned} \hat{s}'([p], xa) &= \delta'(\hat{s}'([p], x), a) && [\text{Defn of } \hat{s}'] \\ &= \delta'([\hat{s}(p, x)], a) && [\text{Induction hypothesis}] \\ &= [\delta(\hat{s}(p, x), a)] && [\text{Defn. of } \delta'] \\ &= [\hat{s}(p, xa)] && [\text{Defn of } \hat{s}] \end{aligned}$$

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Lemma 2. $p \in F$ iff $[p] \in F'$.

Theorem. $L(M/\approx) = L(M)$.

Proof. For $x \in \Sigma^*$,

$x \in L(M/\approx)$ iff $\hat{S}'(\delta', x) \in F'$

iff $\hat{S}'([\delta], x) \in F'$ [Defn. of δ']

iff $[\hat{S}(\delta, x)] \in F'$ [Lemma 3]

iff $\hat{S}(\delta, x) \in F$ [Lemma 2]

iff $x \in L(M)$ [Defn. of acceptance]